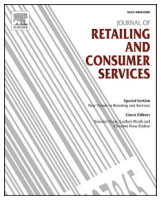
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Do people really want fast and precisely scheduled delivery? E-commerce customers' valuations of home delivery timing

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ABSTRACT

E-commerce marketplaces often offer fast and scheduled delivery services for free to attract customers, imposing strict time constraints on delivery, but do all people really want such services? This study aims to quantitatively analyze heterogeneous customers' preferences for home delivery timing. We propose two willingness-to-pay measures, namely, the value of delivery time savings (VODT) and the value of time slot shortening (VOTS). We capture their distributions in a data-oriented manner by estimating an extended mixed logit model with a semi-nonparametric approach. The estimated VODT ranged from -47.9 to 219.4 JPY/day, with a median of 25.6 JPY/day. This result shows a large taste variation and that most customers were not willing to pay even 10% of the delivery fee (set between 300–600 JPY) to save the waiting time by one day. Moreover, VOTS was found to be low, distributed with a median of 5.0 JPY/hour. The fact that some customers do not necessarily highly value the reduction in delivery lead-time or time slot size sheds light on demand management to reduce delivery burden while maintaining customer satisfaction. In addition, the valuation varied according to the category and price of the purchased good as well as the e-shopping frequency of customers, which suggests a possibility of differentiated pricing or real-time design of service attributes as a retailing strategy.

1. Introduction

E-commerce has experienced rapid growth over the past two decades, significantly changing consumer shopping behavior. Today, many people enjoy online shopping that frees them from travel costs and provides access to an extensive product range along with the option of shopping at home 24/7 (Farag et al., 2007; Hsiao, 2009), and the COVID-19 pandemic has boosted and rather necessitated this transition (UNCTAD, 2021; Laudon and Traver, 2023). The digital transformation associated with e-commerce has dramatically increased the demand for last-mile parcel delivery.

Not only does demand increase quantitatively but experienced customers further desire fast, cost-effective, and precise (within a specified narrow time slot) delivery. To meet such customers' needs, e-marketplaces often provide customer-oriented services, offering their customers fast or scheduled delivery options for free (e.g., Klapp et al., 2020) in reaction to their unwillingness to pay extra for such services (Savelsbergh and Van Woensel, 2016; Rai et al., 2019). However, on the other side, the fulfillment of customers' increasing needs for delivery imposes an excessive burden on logistics (e.g., Hua et al., 2010): (1) fast delivery reduces a *lead-time*, the time required to complete a delivery process, and makes the delivery schedule tight, and (2) precisely scheduled delivery adds strictly *narrow time-window constraints* and thus significantly reduces the routing efficiency of last-mile delivery (Campbell and Savelsbergh, 2005; Boyer et al., 2009; Agatz et al., 2011). As a result, the workload of delivery workers has been intensifying and is recognized today as one of the major social issues faced by many countries (e.g., Ogawa, 2017; Sang-Hun, 2020; Gurley, 2020). While the optimization of costs in different stages has been the primary focus of the literature, customers' behavior is one of the dominant factors affecting the entire delivery process, and therefore, understanding their behavior and preferences can be a key to solving many logistics challenges (Neslin et al., 2006; Holguín-Veras et al., 2017; Nguyen et al., 2019).

In the context of recent e-marketplaces and home delivery, a customer is often provided a set of options for *delivery timing* when (or

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after) purchasing a good. The customer then chooses a timing option considering the delivery attributes such as fee, timing precision and convenience, as well as his/her availability for the receipt. For instance, *Amazon.co.jp* offers three options of home delivery timing with a customer buying a good: (1) expedited, (2) scheduled, and (3) standard shipping.[1] Expedited delivery includes next-day or same-day delivery. Scheduled delivery allows a customer to specify the date and time slot for receiving the ordered good. Standard (or normal) delivery is an option in which a delivery provider can decide on which day and what time of the day to deliver the good, from a set of reasonable (i.e., not too unsatisfactory) timings, thereby being the most flexible option for the supply side. However, in many cases such as a customer holding a privilege (e.g., Amazon Prime) or a retailer doing a campaign, fast or scheduled delivery options are offered for free or set as a default choice. In such a situation, a customer does not have incentives to choose normal delivery and easily chooses fast or scheduled delivery, possibly even if s/he does not want it.

The customers' choices of fast and scheduled delivery impose strict time constraints on delivery. From a managerial perspective, it is necessary to relax such constraints by shifting their choices to a more flexible delivery timing option (with a long lead-time, wide time-window, or sparse demand). While previous marketing studies on e-commerce have suggested the importance of shorter delivery time for sales, customer satisfaction and choice behavior (Marino et al., 2018; Ma, 2017; Nguyen et al., 2018), customers consider the trade-off between convenience (e.g., time and precision) and price of offered delivery options (Kollmann et al., 2012; Agatz et al., 2021). Based on this trade-off concept, several studies have estimated customers' valuations of delivery service attributes (Hsiao, 2009; Gawor and Hoberg, 2019; Amorim et al., 2022). Yet, there is still a gap in research quantitatively analyzing the trade-off of delivery attributes in customer preferences, and the question of profitability and demand management often remains unanswered (Gawor and Hoberg, 2019). Moreover, preferences and sensitivity to service attributes can largely vary across customers (Nguyen et al., 2019; Rai et al., 2019), and some customers may not prioritize speed or precision due to scheduling flexibility or constraints. If we could identify the variation of preferences across customers and then incentivize some customers to a flexible delivery timing option, it would be possible to reduce the delivery burden while maintaining customer satisfaction. How to achieve this balance would contribute to the theory and practice of marketing for e-commerce delivery.

As such, the research question this study aims to answer is "*Do (all)* people really want fast and precisely scheduled delivery?", in other words, "*in which conditions and to what extent such options are needed?*" To this end, this study analyzes customers' preferences in a choice context among different delivery timing options, by performing stated choice analysis based on random utility maximization (RUM) theory (McFadden, 1986). In particular, we invoke the theory of *valuation of time* in transportation economics (Small, 2012) and calculate two willingness-to-pay (WTP) measures called the *value of delivery time savings* (VODT) for speed and the *value of delivery time slot shortening* (VOTS) for timing precision, respectively. The former indicates how much additional delivery fee customers are willing to pay to save their waiting time for their orders by one day, and the latter indicates how much they are willing to pay to shorten a time slot for the receipt by one hour. Because customers do not know exactly when the ordered good is delivered within a time slot, the uncertainty of the delivery timing increases with the time slot size, whose effect is analyzed through VOTS. In this analysis, the question of customer variation is reduced to capturing the interpersonal heterogeneity of VODT and VOTS. However, there is little prior knowledge or studies of such valuations in the e-commerce delivery context. This study therefore develops a data-oriented approach that estimates the distributions of VODT and VOTS without prior knowledge to test the following hypothesis: E-commerce customers' preferences for home delivery timing are largely heterogeneous, and some customers do not highly value the delivery time saving or time slot size shortening in monetary terms.

For the analysis, we perform a stated preference (SP) survey where a respondent is offered a set of delivery option timings with different levels of delivery attributes, while the real situations of current e-marketplaces in which *too* customer-oriented services offered prevent the understanding of the trade-off between the attributes. Also, we observe the detailed characteristics of customers, including demographic and household characteristics, lifestyle and e-commerce experience to take into account the heterogeneity. To analyze the SP data, we estimate a mixed logit (MXL) model (Train, 2009), which is an advanced random utility approach that accommodates taste variation and flexible substitution patterns and has been successfully employed in stated choice analyses. Furthermore, as a data-oriented approach to the estimation of the distributions of VODT and VOTS, we introduce a flexible semi-nonparametric approach by Fosgerau and Mabit (2013).

Again, although fast and scheduled delivery is generally preferred by customers, in the opposite sense, such a customer-oriented option imposes strict constraints on last-mile delivery. The concepts of VODT and VOTS enable us to quantitatively analyze the monetary value a customer is willing to pay for the increase/decrease in the delivery time and time slot size, thereby being helpful in designing pricing strategies. The study also methodologically contributes to the literature by capturing the heterogeneity of the valuations in a data-oriented manner. Given that the recent technological advancement enables personalization or real-time design of service attributes (Waßmuth et al., 2023), the analysis of customers' heterogeneous preferences contributes to the marketing theory and practice of e-commerce delivery for reducing delivery operation costs while maintaining customers' utilities.

The remainder of this paper is structured as follows. Section 2 provides the literature review, along with the positioning and contributions of this study. Section 3 describes the details of the stated choice experiment design. Section 4 reports the data collection and sample statistics. Section 5 introduces the MXL model of delivery timing choice behavior to analyze the stated choice data, and Section 6 presents the estimation results. Finally in Section 7, we conclude the study.

2. Literature review

This section provides the literature review. While an extensive body of literature has investigated e-commerce customers' satisfaction, retention, and loyalty (e.g., Rao et al., 2011; Koufteros et al., 2014; Uzir et al., 2021) or attitude to sustainable delivery options (e.g., Punel and Stathopoulos, 2017; Yuen et al., 2018), a choice-based analysis of customers' preferences provides insights on how customers change their choices in reaction to service design, thereby evaluating various behavioral indicators that are the key to demand management (Ben-Akiva and Lerman, 1985). Given that fast and scheduled delivery is a generally-adopted option in existing e-marketplaces and imposes significant constraints on last-mile parcel delivery, this review mainly focuses on (1) demand management based on customers' choice behavior of delivery timing and (2) analysis of WTP in the delivery demand management context.

2.1. Delivery timing choice analysis

A delivery timing choice model analyzes how a customer chooses a delivery timing, i.e., when s/he receives the ordered item at home. A time slot choice model is a typical case and is often studied in the context of attended home delivery (Waßmuth et al., 2023). By explicitly modeling customers' time slot choice behavior, a number of papers studied the pricing and allocation of delivery time slots (Agatz et al.,

[1] Source: https://www.amazon.co.jp/gp/help/customer/display.html?nodeId=201911210&language=en_US.

2011, 2013; Yang et al., 2016; Yang and Strauss, 2017; Mackert, 2019; Klein and Steinhardt, 2023), where *pricing* determines the delivery fee at which each time slot is offered, while *allocation* deals with which time slots to offer (Klein et al., 2019). As such, the modeling of customers' choice of delivery timing can help the design of delivery demand management strategies by predicting the change in individual customers' choice propensities in reaction to the strategies.

As for the analysis of delivery timing preferences, Agatz et al. (2021) explored the impact of green labels and delivery fees on time slot choice behavior and found that green labels worked as an incentive for eco-conscious people to change their time slot choices. Yang et al. (2016) and Yang and Strauss (2017) estimated time slot choice models in more usual settings but included only the sensitivity to slot price and slot dummy effects in the utility function, thus cannot be able to analyze the trade-off between the delivery fee and other attributes. Recently, Amorim et al. (2022) analyzed customers' preferences for the attribute changes of delivery time slot alternatives, using a large-scale empirical dataset from a grocery retailer, and found that customers not only value speed as a delivery attribute but also consider time slot size to be a key driver of their choice. This analysis was later extended by Ali and Amorim (2023) to take personalization into account.

As such, delivery timing choice analysis provides a better understanding of customer preferences for home delivery timing. However, due to the recent growth of e-commerce and technology, demand management has only started to attract attention and has yet to be sufficiently studied in the literature (Waßmuth et al., 2023). Therefore, previous studies on delivery timing choice analysis have several important limitations. Firstly, most studies solely analyze the choice of a time slot for scheduled delivery. A delivery timing flexible for last-mile delivery, such as normal delivery without a specified time slot, has not been integrally considered with the time slot choice. In addition, many studies used the MNL framework for modeling time slot choice behavior. While the MNL framework provides a demand forecast consistently with the RUM theory, it cannot capture the underlying correlation due to the independence of irrelevant alternative (IIA) property (Ben-Akiva and Lerman, 1985), which is a significant drawback, particularly for a case with many alternatives like time slot choice analysis (Ali and Amorim, 2023). Furthermore, the studies on time slot choice analysis did not sufficiently consider customers' heterogeneity, although the preferences can be heterogeneous among customers. In an empirical setting, a company often does not provide customers' information due to privacy issues, and it is almost impossible to observe the actual set of available time slots and their attributes offered to customers. Amorim et al. (2022) retrieved the available alternatives and their attributes for each customer by a heuristic algorithm, but the uncertainty contained during the processing and possible resultant biases in the results cannot be ignored. Therefore, to further investigate customers' heterogeneity in detail, studies based on an SP survey with a sufficient sample size in which the analyst can directly observe the actual choice set and test different levels of attributes are needed.

2.2. Willingness-to-pay analysis in delivery demand management

A few studies analyzed WTP measures for delivery attribute changes, as summarized in Table 1. VODT has been analyzed in various contexts and various markets. However, although the width of the time window is a main constraint of delivery route planning (Campbell and Savelsbergh, 2005; Agatz et al., 2011), only a few studies analyzed its value on the demand side, i.e., VOTS (Amorim et al., 2022).

As for the choice context, most of the studies analyzed the VODT in the context of *shopping mode choice* (Hsiao, 2009; Schmid and Axhausen, 2019; Meister et al., 2023). They modeled a choice between online and physical in-store shopping and calculated the VODT as the trade-off between delivery time and travel cost for in-store shopping. Similarly, Gawor and Hoberg (2019) analyzed VODT in the context of choice among e-retailers offering the same product with different prices (including delivery fees) with different delivery options. In these contexts, the VODT is not calculated purely between delivery fee and time, and this measure cannot be directly used for designing a pricing scheme for different delivery options.

In contrast, Amorim et al. (2022) analyzed VODT and VOTS in the context where multiple time slots with different fees, speeds, and time slot sizes are offered, which can help with the pricing of time slots. Amorim et al. (2022) also estimated an MXL model but did not find unobserved heterogeneity in customers' price sensitivity. For capturing the heterogeneity, a main drawback of their study is that they relied on revealed preference (RP) data that did not identify individuals and capture their repetitive choices. Also, they assumed the random coefficients of the attributes that follow a normal distribution. Earlier studies in transportation research give cautions on its use because the strong assumptions of a normal distribution often lead to an unrealistic WTP distribution that can not be revealed by data (e.g., Hess et al., 2005). To better capture inter-personal heterogeneity, panel data including multiple observations from each individual is needed (Train, 2009). Furthermore, Train and Weeks (2005) and Fosgerau (2007) pointed out that choice models constructed in preference space may provide less reasonable distributions of WTP than the models in WTP space for environmental and transportation applications.

In addition, all the reviewed studies on VODT and VOTS focus on a specific single market. Therefore, their results lack generalizability among different markets.

2.3. Contributions of the study

In this study, we significantly enhance the analysis of VODT and VOTS by designing and implementing the stated choice analysis regarding e-commerce customers' delivery timing choices. With a particular focus on customers' heterogeneity, we reveal under which conditions and to what extent people would need fast and precisely scheduled delivery. The contributions of this study are summarized as follows:

- **Choice context:** We analyze the VODT and VOTS in the context of the integrated choice of delivery timing, including an option choice among fast, scheduled and normal delivery, and a time slot choice when scheduled delivery is chosen. This choice context has not been sufficiently studied in the literature but represents a more realistic situation that e-commerce customers often face when shopping online, e.g., in Amazon. The novelty of this analysis lies in the evaluation of the trade-off between the attributes purely associated with delivery options, which allows us to derive the monetary values of the plain changes in delivery speed and precision. The integrated choice analysis also gives us an insight into how to incentivize customers to select a flexible option for suppliers against scheduled delivery with specified time slots. Also, unlike the literature, our analysis is not limited to a single market but includes various types of market items, thereby obtaining a generalizable result.
- **Stated choice analysis:** We designed an SP survey to observe multiple responses from each individual and his/her personal characteristics. Moreover, we performed the survey in collaboration with the largest company in Japanese parcel delivery, inviting their customers as survey respondents. This enabled us to collect a rich, large, and representative sample to address the limitations of the previous studies on capturing heterogeneity, such as lack of personal information in RP data (Amorim et al., 2022) and of sample size (e.g., Zhang and Huang, 2022; Nguyen et al., 2019).
- **Capturing heterogeneity:** We capture the heterogeneity in customers' preferences in detail based on the large-scale stated choice data, while only the single value (point estimate) of VODT was often estimated in the literature (Hsiao, 2009; Dinlersoz and Li, 2006; Gawor and Hoberg, 2019). We estimate the distributions of VODT and VOTS by using an extended MXL model. The results re-

Table 1
Willingness-to-pay analysis in delivery demand management.

Author(s)	Context	VODT	VOTS	Market	Data	Model	Heterogeneity
Hsiao (2009)	Shopping mode choice	✓	x	Book	SP	Binary logit	x
Gawor and Hoberg (2019)	Retailer choice	✓	x	Electric	SP	Conjoint	x
Schmid and Axhausen (2019)	Shopping mode choice	✓	x	Grocery	SP	Hybrid choice[a]	✓
Meister et al. (2023)	Shopping mode choice	✓	x	Grocery	SP	Hybrid choice	✓
Amorim et al. (2022)	Time slot choice	✓	✓	Grocery	RP	Mixed logit	△[b]
This study	Integrated timing choice[c]	✓	✓	Various	SP	Mixed logit	✓

[a] It is also called an integrated choice and latent variable (ICLV) model.
[b] Heterogeneity is limitedly considered with respect to characteristics of the basket but not of the customer.
[c] Integrated choice of delivery options (fast, scheduled and normal) and time slots (if scheduled delivery is chosen).

vealed significant inter-personal heterogeneity in VODT and VOTS, implying that there are some customers who do not really need fast or precisely scheduled delivery and would change their choices even with a small additional shipping fee. We also found systematic heterogeneity with respect to the ordered item price and online shopping frequency.

- **Model development:** We also contribute to the literature on delivery timing choice models in terms of model development. We integrate a semi-nonparametric approach by Fosgerau and Mabit (2013) into the MXL framework, which allows us to flexibly capture the distributions of VODT and VOTS without relying on any strong distributional assumption such as normal distribution (Amorim et al., 2022). Moreover, we capture complex substitution patterns that the alternatives of the integrated choice may exhibit, by incorporating error components into the MXL model with a cross-nested structure. As such, this study fills the gap in modeling technique as many studies of delivery timing choice relied on the standard MNL framework suffering from the IIA property (e.g., Yang et al., 2016; Agatz et al., 2021).

3. Experimental design

We designed a web-based SP survey for home delivery timing choices of e-commerce customers. The survey consisted of three parts: (1) questions on the most recent online shopping experience, (2) stated choice tasks on delivery options, and (3) questions on socio-demographic characteristics, and household and lifestyle attributes. Note that the first part of the survey was conducted to establish a realistic situation for the respondents during the stated choice tasks.

3.1. Stated choice task

For the stated choice experiment, we asked respondents to assume that they order the same item at the same price as those ordered in their most recent online shopping transaction. The date on which they responded to the survey was assumed to be the order date for the stated choice tasks.

Fig. 1 shows an example of the stated choice task. For each task, the respondent first chooses a delivery option among “next-day”, “scheduled” and “normal” delivery (Q1), where normal delivery is defined accordingly to standard delivery of Amazon, which gives flexibility to a shipper to choose the date and time for delivery. Respondents who chose scheduled delivery were asked to jointly select the date and time slot for delivery (Q2 and Q3). The available dates for scheduled delivery spanned from 2 to 8 days after the order date (notated as “2+” ∼ “8+”). In Q2, the respondent specified the order date on a calendar popping up on click. While the entire period of delivery hours in a day was fixed during 9:00–21:00 (12 hours in total), the size of a time slot varied from 2 to 4 hours by choice scenarios, resulting in from six to three available slots. Fig. 1 shows a case where the time slot size is 3 hours, and therefore, the number of available slots is four. For next-day delivery, the ordered item was assumed to be delivered on the following day, but the respondent was not allowed to select a time slot. When respondents chose normal delivery, they could neither select a date nor a time slot for delivery.

Note that this integrated choice task of delivery option, date, and time slot is a realistic choice problem that e-commerce customers often face when shopping online, e.g., in *Amazon*. Nevertheless, existing literature has focused only on a single aspect, such as delivery date (speed) (e.g., Rai et al., 2019; Gawor and Hoberg, 2019) or time slot choice (e.g., Agatz et al., 2021; Amorim et al., 2022).

3.2. Attributes

The delivery attributes on which this study mainly focused are fee, time (speed), and the size of a time slot. To characterize the delivery timing alternatives, the eight attributes were controlled in the stated choice tasks, as summarized in Table 2.

The delivery fees were differentiated by delivery options, ranging from 300 to 600 JPY (Attributes 1–3).[2] For scheduled delivery, an additional charge of 100 JPY was considered for delivery on holidays and in the latest time slot (Attributes 4–5), respectively.

The expected delivery time for normal delivery is based on the earliest possible date (Attribute 7) and the range of delivery dates (Attribute 8). The delivery time for next-day delivery is always one day, and that for scheduled delivery directly depends on the choice of the respondents, ranging from 2 to 8 days.

As mentioned in the previous subsection, the size of a time slot is an attribute for scheduled delivery and varies from 2 to 4 hours (Attribute 6). The entire delivery period in a day ranges from 9:00 to 21:00, with the time slots defined by the slot size. The number of available slots, therefore, can be three, four, or six. When the slot size is three hours, for example, the available slots are [9–12, 12–15, 15–18, 18–21] (as in Q3 of the example in Fig. 1).

In total, the experiment had 3 four-level, 2 three-level, and 3 two-level attributes, implying $4^3 \times 3^2 \times 2^3$ profiles for a full factorial design, which is very hard to manage. Therefore, we created a fractional factorial design consisting of 400 profiles where the main effects were orthogonal to each other, using the support.CE package for R (Aizaki, 2012). From this set of choice tasks, we randomly displayed five tasks to each respondent. Note that we did not consider optimal designs like D-efficient designs (Kuhfeld et al., 1994) because prior values of the parameters for the delivery option choice model were not empirically known. Moreover, Walker et al. (2018) recently presented that random design, despite its simplicity, performs as well as any design in terms of robustness.

[2] The values were selected considering the current shipping rates of Amazon.co.jp: 410–450 JPY for normal delivery, 510–650 JPY for fast delivery, and 510–550 JPY for scheduled delivery (Source: https://www.amazon.co.jp/gp/help/customer/display.html?nodeId=201911210&language=en_US).

	(1) Next-day Delivery	(2) Scheduled Delivery (From 2 to 8 days after)	(3) Normal Delivery (Date or time can not be specified)
Delivery Fee	***600 JPY***	***500 JPY***	***300 JPY***
Additional Charge	None	***Holiday: + 100 JPY*** ***Night : + 100 JPY***	None
Delivery Date	Next day	Specified date	***3 ~ 5 days after*** (not specified)
Time slot	Not available	***Select from 4 slots of 3 hours***	Not available

Q1. Which delivery option would you choose?

◯ (1) Next-day Delivery

◯ (2) Scheduled Delivery (from 2 to 8 days after)

◯ (3) Normal Delivery (specified day or time are not available)

Q2. If you chose "(2) Scheduled Delivery," please select your preferred delivery date from the calendar below.

Note: please consider today the order date.

Calendar pops up on click

Q3. If you chose "(2) Scheduled Delivery," please select your preferred delivery time slot from the list below.

◯ ***9:00 - 12:00***

◯ ***12:00 - 15:00***

◯ ***15:00 - 18:00***

◯ ***18:00 - 21:00***

Fig. 1. Example of the stated choice task: Items in bold and italic indicate designed task attributes. This is an English translation of the survey (see Fig. A.7 for the original).

Table 2
Attributes and attribute levels.

	Attribute	Alternative	No. of levels	Values
1	Delivery fee (JPY)	Next-day	4	[300, 400, 500, 600]
2	Delivery fee (JPY)	Normal	4	[300, 400, 500, 600]
3	Delivery fee (JPY)	Scheduled	4	[300, 400, 500, 600]
4	Additional charge for holiday (JPY)	Scheduled	2	[±0, +100]
5	Additional charge for the latest slot (JPY)	Scheduled	2	[±0, +100]
6	Size of a slot (hrs)	Scheduled	3	[2, 3, 4]
7	Earliest possible date (d+)	Normal	2	[2+, 3+]
8	Range of delivery dates (days)	Normal	3	[2, 3, 4]

4. Stated choice data

4.1. Data collection

We conducted the stated choice survey from April 30 to May 14, 2021, during the COVID-19 pandemic, collaborating with Yamato Holdings Co., Ltd. which has the largest market share of parcel delivery in Japan (42.3% in the Financial Year 2018). The respondents were among the Kuroneko Members (more than 50 million in Japan), registered customers of Yamato Holdings' delivery service. In this survey, we randomly selected customers who live within the three major metropolitan areas (Greater Tokyo, Osaka, and Nagoya) in Japan. The population of the three areas comprises 52 million, accounting for 41% of Japan's total population.[3] The survey invitation was sent to randomly sampled 100,000 Kuroneko Members, of whom 4,872 completed it (i.e., the response rate was 4.87%). Since five choice tasks were given to each respondent, the original sample size observed was 24,360. Statistics on

[3] Source: 2015 Population Census (Statistics Bureau, Ministry of Internal Affairs and Communications) https://www.stat.go.jp/english/data/kokusei/2015/summary.html.

Japanese e-commerce users and Kuroneko Members as a whole are not publicly available, but given that almost half of the Japanese population registered for the membership, a random sampling from Kuroneko Members would be sufficiently representative of the target population.

4.2. Data cleaning

We cleaned the data in the following way:

- First, we removed 635 observations where the scheduled delivery option was chosen for the next day or after more than 8 days.[4] As mentioned earlier, we accepted the scheduled delivery only for 2-8 days after the order date. This cleaning reduced the sample size from 24,360 to 23,725.
- Then, we removed 745 observations of 151 respondents who have never experienced online shopping. To analyze the preferences for delivery options in e-commerce, this study focuses on experienced users.[5] Afterward, the sample size was reduced to 22,980.
- Finally, we removed 3120 observations of 655 respondents who chose once or more an alternative dominated by other options. Such alternatives include the normal delivery option with higher delivery fees than the next-day or scheduled delivery options.[6] As a result, the sample size became 19,860.

After data cleaning, we finally obtained 19,860 observations of 4,062 unique respondents.

4.3. Sample statistics

Table 3 lists the descriptive statistics of the demographic characteristics, household information, and e-commerce attributes of the respondents, as well as their recent online shopping transactions. Because of the COVID-19 pandemic, 26.7% of the respondents telecommute (work from home) once a week or more. In addition, 25.3% of the respondents have a delivery box at their places, which is a locker-type device where parcels can be left even when the recipient is not at home.

Regarding online shopping experiences, 21.6% of the respondents usually buy an item online once a week or more, and 46.1% of the respondents have registered for a membership with a free delivery privilege. Note that it was clearly explained in the stated choice experiment that all respondents were required to pay the delivery fee even if they have such membership, and this variable was collected mainly for testing the inertia of customers' choice behavior.

We observed a wide range of variety in category and price of items purchased. The information regarding the item price has been used in analyzing the WTPs, which is important given that the literature targeting different marketplaces reported significantly different values of WTP for delivery (Hsiao, 2009; Gawor and Hoberg, 2019; Schmid and Axhausen, 2019; Meister et al., 2023).

4.4. Aggregation

Out of 19,860 observations, the next-day, normal and scheduled delivery options were chosen 8,429 (42.4%), 4,265 (21.5%), and 7,166 times (36.1%), respectively. Of scheduled delivery, weekdays and holidays were chosen 1,327 (18.5%) and 5,839 times (81.5%), respectively; and morning, daytime, and night time slots were chosen 3,262 (45.5%), 2,082 (29.1%) and 1,822 times (25.4%), respectively, where the morning and night time slots are defined as the earliest and latest time slots, and the other slots are considered as daytime slots. The reason why the night time slot was least demanded is that we randomly imposed an additional charge of 100 JPY on the night time slots. Indeed, regarding the observations where the additional charge was not imposed, the morning, daytime, and night slots were chosen 1,533 (41.6%), 812 (22.0%), and 1,338 times (36.3%). This also shows a pricing scale in the Japanese home delivery context, and 100 JPY is effective to shift customers' choices.

Although next-day delivery was the most popular alternative, the other options were also chosen a considerably large number of times. To see the trade-off between delivery time and fee, we plot the aggregate choice results focusing on the two attributes in Fig. 2, which includes the results of next-day (the delivery time is one day) and scheduled delivery (two to eight days). Since a delivery date was not fixed for normal delivery, this aggregation does not include the observations choosing it. Fig. 2(a) shows the choice results based on the delivery time and fee that each option offers. Most of the respondents seem to have preferred delivery within three days after the order date. There is no major change in the number of choices from the fourth to the eighth day after the order date, which may be because, for deliveries later than three days after, customers cared more about their schedule than the delivery time. Delivery within three days seems to be popular with a wide range of delivery fees: even up to JPY600, the highest possible fee for next-day delivery, one to three days after the order date were chosen relatively high numbers of times.

Note that, due to the SP task design, JPY of 500 and 600 were not always expensive among the available alternatives. This can be seen in Fig. 2(b), which focuses on the delivery fee difference to the minimum among the fees of the offered alternatives, instead of the pure delivery fee. It shows that next-day delivery was chosen 6,426 times (32.4% of the total) when it was the least expensive among the available options. That being said, customers consider the trade-off between delivery fee and time. Although some customers still preferred their parcels to be delivered on the next day, many of them changed their preferences when the fee was not the cheapest.

5. Delivery timing choice model

To analyze the stated choice data, we perform a discrete choice analysis of home delivery timing. The alternatives for the choice problem are based on delivery options (next-day, scheduled, and normal), dates, and time slots. As mentioned earlier, the number of available time slots varies by choice scenarios. To analyze all samples in a unified manner, we categorize the time slots into three categories: "morning" (the earliest), "night" (the latest), and "daytime" slots. Note that the consistency in additional charge is retained with this definition (Attribute 5 in Table 2). As such, we have 21 (7×3) time slots for scheduled delivery, resulting in 23 alternatives in the choice set. The choice set is unique and the same for every customer.

5.1. Model formulation

For the analysis, we estimate an MXL model that captures unobserved inter-individual heterogeneity and flexible substitution patterns. The MXL framework is a highly flexible model and has been successfully employed in stated choice analysis. Based on the RUM theory, individual $n \in \{1, \ldots, N\}$ is assumed to choose an alternative that maximizes the utility U_{ntj} in choice task $t \in \{1, \ldots, T_n\}$:

$$U_{ntj} = V(\boldsymbol{X}_{ntj}; \boldsymbol{\beta}_n) + \varepsilon_{ntj}, \tag{5.1}$$

[4] We could not restrict the available order dates for each participant due to system specifications. Alternatively, we carefully explained that the available dates for the scheduled delivery were 2 to 8 days after the order date (i.e., survey date).

[5] This processing is also done in Nguyen et al. (2019), who also focused on respondents who have at least some experience shopping online.

[6] This study considered these responses irrational, and to detect respondents who make such a choice, we purposely included choice tasks with dominated alternatives in the profiles.

Table 3
Sample statistics ($N = 4062$).

Category	Characteristic	Value	Sample frequency
Socio-demographic	Age	≤ 29	2.8%
		30–49	29.9%
		50–69	58.8%
		≥ 70	8.5%
	Gender	Female	41.3%
		Male	58.7%
	Occupation	Employee	44.1%
		Self-employed	5.2%
		Public officer	4.6%
		Professional (including healthcare worker)	3.4%
		Part-time job	12.5%
		Student	1.0%
		No job	20.0%
		Other	9.2%
	Telecommuting frequency	Everyday	7.8%
		3–4 days per week	9.2%
		1–2 days per week	9.7%
		2–3 days per month	4.1%
		1 day per month or less	11.1%
		Never	58.1%
Household	Residence	Tokyo	29.6%
		Other	70.4%
	Household composition	Live alone	19.1%
		Couple with double income	17.0%
		Couple with single income	14.9%
		Live with children	32.9%
		Live with family (w/o children)	11.4%
		Live with friends	0.5%
		Other	4.2%
	House type	Detached house	49.1%
		Apartment	49.5%
		Other	1.4%
	Have delivery box?	Yes	25.3%
		No	74.7%
E-commerce attributes	E-shopping frequency	3–4 times per week	3.0%
		1–2 times per week	18.6%
		2–3 times per month	42.3%
		Once per month or less	36.1%
	E-commerce membership	Member w/ free-delivery privilege	46.1%
		Member w/o privilege	48.8%
		Not a member	5.1%
Recent e-shopping experience	Category of item	Books	6.9%
		Software	1.0%
		Computer devices	8.7%
		Grocery	18.9%
		Clothing	15.0%
		CD / DVD	2.9%
		Gift	1.8%
		Daily necessities	6.6%
		Toys	2.2%
		Electric devices	10.5%
		Healthcare goods	6.1%
		Cosmetics	6.2%
		Office supplies	1.5%
		Interior items	2.1%
		Other	9.6%
	Price of item (JPY)	≤ 999	3.9%
		1,000–2,999	20.7%
		3,000–4,999	20.3%
		5,000–7,499	12.4%
		7,500–9,999	11.2%
		10,000–14,999	10.4%
		15,000–19,999	6.1%
		20,000–29,999	4.9%
		30,000–49,999	4.9%
		≥ 50,000	5.2%

where V is the systematic utility, a function of observed explanatory variables $\boldsymbol{X}_{ntj}$, and ε_{ntj} is the unobserved utility following extreme value distribution Type I, i.e., $\varepsilon_{ntj} \sim \text{Gumbel}(0, \mu)$. We assume that the scale μ is standardized as one. The preference parameters $\boldsymbol{\beta}_n$ of individual n are assumed to be realizations from distribution $f(\boldsymbol{\beta}|\boldsymbol{\Omega})$ where $\boldsymbol{\Omega}$ is a set of hyperparameters. When $\boldsymbol{\beta}_n$ is given, the probability that choice y_{nt} of individual n in choice scenario t is j is given by the MNL model (also called the *logit kernel*):

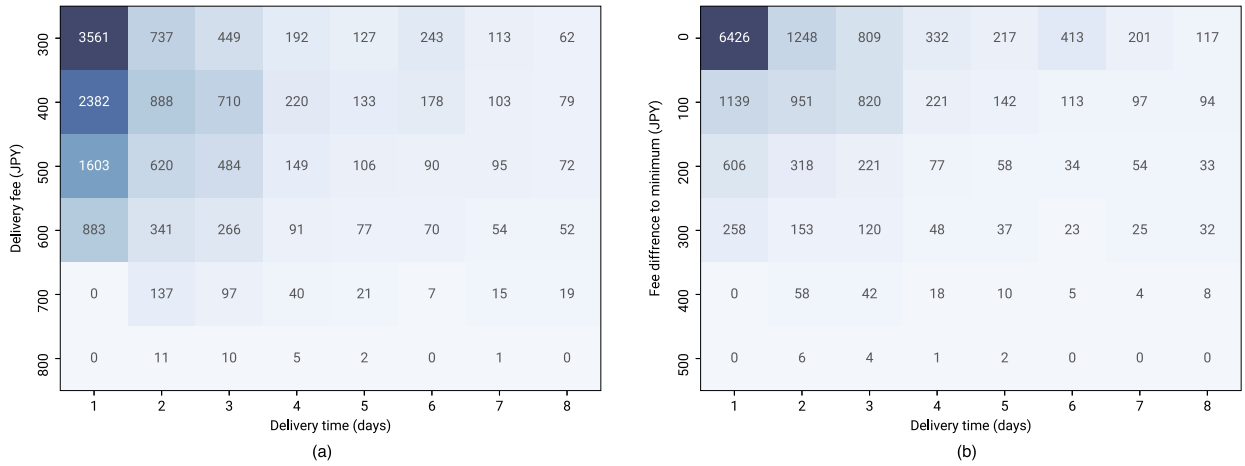


Fig. 2. Choice results with the focus on the delivery time (horizontal axis) and delivery fee (vertical axis): (a) takes a pure delivery fee, and (b) focuses on the fee difference among the offered alternatives. The numbers in the grids indicate the number of times chosen, with deeper colors representing higher numbers. Note that this aggregation includes only the observations that chose the next-day or scheduled delivery options, and therefore the total number does not correspond to 19,860, which is the number of the total observations.

$$P_n(y_{nt}=j|\boldsymbol{\beta}_n)=\frac{\exp\{V(\boldsymbol{X}_{ntj};\boldsymbol{\beta}_n)\}}{\sum_{j'\in C_{nt}}\exp\{V(\boldsymbol{X}_{ntj'};\boldsymbol{\beta}_n)\}}, \tag{5.2}$$

where C_{nt} is the choice set for individual n at choice task t.[7] Therefore, the joint probability of the sequence of choices $\{j_t\}_{t=1}^{T_n}$ for the given $\boldsymbol{\beta}_n$ is

$$P_n\left(\boldsymbol{y}_{nt}=\{j_t\}_{t=1}^{T_n}|\boldsymbol{\beta}_n\right)=\prod_{t=1}^{T_n}P_n(y_{nt}=j_t|\boldsymbol{\beta}_n). \tag{5.3}$$

The MXL model marginalizes this probability over the parameter distribution $f(\boldsymbol{\beta}|\boldsymbol{\Omega})$, obtaining the unconditional probability

$$P_n\left(\boldsymbol{y}_{nt}=\{j_t\}_{t=1}^{T_n}|\boldsymbol{\Omega}\right)=\int_{\boldsymbol{\beta}}P_n\left(\boldsymbol{y}_{nt}=\{j_t\}_{t=1}^{T_n}|\boldsymbol{\beta}\right)f(\boldsymbol{\beta}|\boldsymbol{\Omega})\mathrm{d}\boldsymbol{\beta}, \tag{5.4}$$

and the log-likelihood function is given by

$$LL(\boldsymbol{\Omega})=\sum_{n=1}^{N}\ln P_n\left(\boldsymbol{y}_{nt}=\{j_t\}_{t=1}^{T_n}|\boldsymbol{\Omega}\right). \tag{5.5}$$

Because the integral in (5.4) is hard to compute, we estimate the MXL model by maximizing the following simulated log-likelihood function (Train, 2009):

$$SLL(\boldsymbol{\Omega})=\sum_{n=1}^{N}\ln\left(\frac{1}{R}\sum_{r=1}^{R}P_n(\boldsymbol{y}_{nt}=\{j_t\}_{t=1}^{T_n}|\boldsymbol{\beta}_{nr})\right)\approx LL(\boldsymbol{\Omega}) \tag{5.6}$$

where $\boldsymbol{\beta}_{nr}$ is the rth realization ($r=\{1,\ldots,R\}$) of n's preference parameters drawn from distribution $f(\boldsymbol{\beta}|\boldsymbol{\Omega})$ and R is the total number of random draws.

5.2. Model specification

5.2.1. Utility function

This study investigates the distributions of two WTP measures for delivery, VODT w_d and VOTS w_h, as well as the effects of customers' characteristics $\boldsymbol{s}$ on delivery timing choices. Therefore, instead of taking the ratio of marginal utilities of the target attributes and the delivery fee (i.e., modeling in the preference space), we formulate the model in the WTP space (Train and Weeks, 2005), leading to a straightforward estimation of the WTP distributions. With this formulation, the WTP distribution is directly estimated in a stable manner without any restriction on the distribution of the delivery fee coefficient β_c (Scarpa et al., 2008). The utility U_{ntj} for alternative j of individual n in choice situation t is:

$$U_{ntj}=\beta_{c,n}(\mathrm{DF}_{ntj}+w_{d,n}\mathrm{DT}_{ntj}+w_{h,n}\mathrm{SS}_{ntj})+\boldsymbol{\alpha}'\boldsymbol{s}_{nj}+\boldsymbol{\delta}_j\boldsymbol{\xi}+\varepsilon_{ntj} \tag{5.7}$$

where DF_{ntj}, DT_{ntj}, and SS_{ntj} are the delivery fee (JPY), expected delivery time (days), and time slot size (hours), respectively; DT_{ntj} equals 1 for next-day delivery, 2–8 for scheduled delivery, and is defined as $\mathrm{ED}_{ntj}+\mathrm{RD}_{ntj}/2$ for normal delivery, where ED_{ntj} is the days to the earliest possible date and RD_{ntj} is the range of delivery dates (see Table 2); SS_{ntj} is an attribute associated with scheduled delivery and thus is fixed to be zero for next-day and normal delivery. The parameter $\boldsymbol{\alpha}$ is a vector of the coefficients for the customers' characteristics. The normal error components $\boldsymbol{\xi}=\{\xi_m\sim\mathcal{N}(0,\sigma_m^2)\}_{m\in M}$ capture the heteroskedasticity and correlations among utilities of different alternatives (Train, 2009), with a nesting structure M, and δ_{jm} equals one if alternative j belongs to nest m and zero otherwise (see Section 5.2.4 for the detail). We also assume that β_c, w_d and w_h are randomly distributed parameters.

5.2.2. Observed heterogeneity

We capture the nonlinear observed heterogeneity across individuals, by incorporating interactions of the WTPs with customers' characteristics. By following the continuous interaction specification taken in the estimation of the value of travel time savings (VTTS) (Axhausen et al., 2008), the delivery fee coefficient $\beta_{c,n}$ is structured by the price of item PI_n that user n ordered[8]:

$$\beta_{c,n}=\hat{\beta}_{c,n}\left(\frac{\mathrm{PI}_n}{\mathrm{PI}_{\mathrm{ref}}}\right)^{\gamma_{\mathrm{PI}}}, \tag{5.8}$$

and VODT $w_{d,n}$ is structured by the frequency of online shopping FR_n,

$$w_{d,n}=\hat{w}_{d,n}\left(\frac{\mathrm{FR}_n}{\mathrm{FR}_{\mathrm{ref}}}\right)^{\gamma_{\mathrm{FR}}}, \tag{5.9}$$

[7] This study assumes that C_{nt} is always the same for all choice tasks of all individuals, i.e., $C_{nt}=C$, and consists of 23 alternatives.

[8] This is to capture the difference in sensitivity to the delivery fee depending on the item price. Gawor and Hoberg (2019) reported such a difference between electronic and book marketplaces by comparing their results with those of Dinlersoz and Li (2006) and Hsiao (2009). However, how the item price actually affects the WTPs has never been investigated.

where $\hat{\beta}_{c,n}$ and $\hat{w}_{d,n}$ are realizations from the distributions; $\mathrm{PI}_{\mathrm{ref}}$ and $\mathrm{FR}_{\mathrm{ref}}$ are the sample averages; and γ_{PI} and γ_{FR} are the parameters to be estimated.

5.2.3. Unobserved heterogeneity: distributional specification

We assume that β_c follows a log-uniform distribution, which has a shorter tail than a log-normal distribution and its preferable property has been tested for the value of travel time savings in studies by Fosgerau (2006) and Hess et al. (2017). Therefore, the fee coefficient is given by

$$\hat{\beta}_c = -\exp(a + bu), \tag{5.10}$$

where $u \sim \mathrm{Uniform}(0,1)$, and a and b are the lower bound and the spread, which are the parameters to be estimated.

Unlike the VTTS in the transportation research field, we do not have sufficient empirical evidence for the WTPs of interest. Therefore, assuming the parametric distributional form a prior, such as a normal distribution (Amorim et al., 2022), may bias the understanding. After extensive testing, for the distributional specification of the WTPs w_d, w_h, we apply a semi-nonparametric approach by Fosgerau and Mabit (2013) that does not require a distributional assumption and that can flexibly describe the shape of the distribution in a data-oriented manner. We transform a draw u from a distribution using a power series

$$f(u|\boldsymbol{\eta}) = \sum_{k=0}^{K} \eta_k u^k, \tag{5.11}$$

where K is the dimension of the polynomial expansion. We then compute random draws of WTPs:

$$\hat{w}_d = f(u_d|\boldsymbol{\eta}_d), \tag{5.12}$$

$$\hat{w}_h = f(u_h|\boldsymbol{\eta}_h), \tag{5.13}$$

where $\boldsymbol{\eta}_d = (\eta_{d,0}, \ldots, \eta_{d,K_d})$ and $\boldsymbol{\eta}_h = (\eta_{h,0}, \ldots, \eta_{h,K_h})$ are the vectors of the parameters to be estimated and define the distributional forms of the WTPs of interest, and K_d and K_h are the dimensions of the power series for VODT and VOTS, respectively.

5.2.4. Error components

We have 23 alternatives in the choice set, some of which may introduce correlations violating the IIA assumption of the MNL model that is often adopted in the literature (e.g., Yang et al., 2016; Agatz et al., 2021; Amorim et al., 2022). To capture the heteroscedasticity and underlying correlation among the utilities of delivery timing options, we introduce the error components $\boldsymbol{\xi}$ in the MXL model. Fig. 3 shows the cross-nested structure of the model, where we introduce eight error components, i.e., $|M| = 8$, for the delivery options (next-day, normal and scheduled), as well as date (holiday and non-holiday) and time slot nests (morning, night and daytime). For example, the utility of scheduled delivery two days after the order is placed and in the morning slot of a holiday is defined as:

$$U_{nt,(2+,\mathrm{morning})} = \cdots + \xi_{\mathrm{scheduled}} + \xi_{\mathrm{holiday}} + \xi_{\mathrm{morning}} + \varepsilon_{nt,(2+,\mathrm{morning})}$$

where $\xi_m \sim \mathcal{N}(0, \sigma_m^2)$, $\forall m \in M$, and the standard deviation σ_m is to be estimated. Note that whether "2+" is a holiday depends on the order date for each customer, therefore, $U_{nt,(2+,morning)}$ does not necessarily contain the error component $\xi_{holiday}$. Because we rely on panel data for model estimation, all scale parameters are identified (Walker et al., 2007).

6. Model estimation results

This section reports the model estimation result. The model estimation based on the maximization of simulated likelihood was implemented using the Apollo package in R (Hess and Palma, 2019a,b).

6.1. Final model specification

An extensive specification search has been conducted to reach the final model. Based on our hypotheses and the search process, we mainly focused on the interactions between option choice baselines and individual demographic characteristics and those between slot choice baselines and household and lifestyle characteristics, as well as the delivery attributes. As for day choice, we included the holiday baseline without interaction because holidays were often chosen regardless of the category of respondents.

Table 4 reports the comparison of the models developed during the critical stages of the modeling procedure. First, a standard MNL model was estimated, where no parameters were assumed to be random. Since its framework restricts substitution patterns and does not adequately account for preference heterogeneity across respondents, MXL specifications were then explored. The error components were introduced into the model to capture heteroskedasticity and flexible substitution patterns across the alternatives (MXL 1). Furthermore, three MXL specifications were explored in terms of unobserved preference heterogeneity by assuming the delivery fee coefficient and the two WTPs to be randomly distributed (MXL 2–4). For the distributions of the WTPs, the normal and log-normal distributions were tested (MXL 2 and MXL 3), as well as the polynomial approximation of (5.11) (MXL 4). After a comprehensive trial,[9] the dimensions of the polynomial approximations were finally set at $K_d = 4$ and $K_h = 2$, and uniformly distributed random numbers were used as the base u in (5.11). The overall goodness-of-fit of the models appears to be satisfactory, and the incorporation of both error components and random WTP parameters significantly improved the model performance (Table 4). Consequently, MXL 4 obtained the best goodness-of-fit in terms of the values of log-likelihood, AIC, BIC and $\bar{\rho}^2$. Especially, the polynomial approximations allowed the model to better fit compared to normal and log-normal distributions.

6.2. Estimation results

Table 5 enumerates the estimation results for three models: MNL, MXL 1 (error component model) and MXL 4 (error component and random parameter model with polynomial approximations). In the interest of brevity, we herein do not report MXL 2 and 3 that were outperformed by MXL 4. Overall, the sensitivities to the delivery attributes were estimated with statistical significance and have the expected signs. Most of the other parameters also correspond to our expectations. Below we discuss the results in detail.

6.2.1. Observed heterogeneity: effects of customers' characteristics

We first discuss observed heterogeneity for delivery timing choice preferences, which was captured by interactions between alternative specific constants and customers' characteristics. For the sake of representativeness in the effects of characteristic variables, the categories of some variables were merged before the model estimation. As such, we defined respondents who worked from home for 1-2 days per week or more as *telecommuters* (30.8% of sample, see Table 3). For the subgroups of choices, normal delivery (option choice), weekday delivery (date choice), and daytime slot (slot choice) are treated as references for the alternative specific constants, respectively.

Option choice The estimation result indicates that users who are 70 years or older tend to choose normal delivery compared to next-day and scheduled delivery, while users younger than 30 years old exhibit

[9] We tried two candidates for bases, namely uniform and normal distributions, and different dimensions from two to five for polynomial approximations respectively in (5.12) and (5.13). Considering the resultant distributions, goodness-of-fit, and estimation stability to the settings, we finally selected the base and dimensions reported.

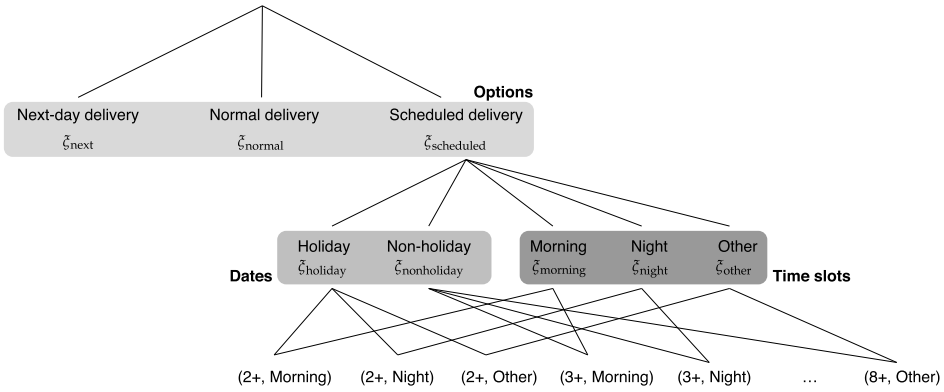


Fig. 3. Cross-nested structure of the choice model approximated by the error components.

Table 4
Model comparison.

	MNL	MXL 1	MXL 2	MXL 3	MXL 4
Error component	No	Yes	Yes	Yes	Yes
Distribution of VODT	Fixed	Fixed	Normal	Log-normal	Polynomial ($K_d = 4$)
Distribution of VOTS	Fixed	Fixed	Normal	Log-normal	Polynomial ($K_h = 2$)
No. of parameters	46	54	57	57	61
Log-likelihood	-35,068.2	-28,359.06	-26,812.02	-26,796.48	-26,624.67
AIC	70,228.4	56,826.12	53,738.05	53,706.96	53,371.35
BIC	70,591.64	57,252.53	54,188.15	54,157.06	53,853.03
$\bar{\rho}^2$	0.4361	0.5437	0.5685	0.5688	0.5715

opposite tendency, suggesting a clear difference in the delivery option preference between different age groups. Additionally, male users prefer next-day delivery but not scheduled delivery, which implies that they tend to order items just before they need them. People living in Tokyo, the largest and busiest city in Japan, prefer scheduled delivery that allows them to specify the delivery date and slot to fit into their schedule. In contrast, the estimation result indicates that telecommuters do not need scheduled delivery, reflecting their higher flexibility in schedule to receive the parcels at home. During the COVID-19 pandemic, many people started working remotely and last-mile delivery may have been released from complex time-window constraints than before. However, non-telecommuters who prefer scheduled delivery and delivery providers may still have a problem due to the rapid growth in total demand for last-mile delivery. Moreover, customers who have a privileged membership in an e-commerce marketplace particularly prefer next-day delivery to normal delivery. Because we clearly explained that their privilege was not valid for the stated choice tasks, this result highlights the inertia of their choices, i.e., they are used to requesting fast delivery. Finally, the result suggests a strong relationship between the presence of a delivery box and customers' delivery timing choice preference. If customers have a delivery box equipped with their homes, they do not need to worry about the timing of receipt of items. As such, they are willing to opt for next-day delivery, thereby, not needing scheduled delivery.

Time slot choice Regarding delivery time slot preference, the analysis focused on its relationships with the characteristics of occupations and households, reflecting customers' lifestyles. First, customers living with their family do not prefer the latest slot delivery, implying that they can afford to receive their parcels during the day, instead of in the evening, when the family spends time together. The time slot choice also depends on housing types: people who live in detached houses tend to choose slots during the day, possibly caring for their neighbors, indicated by the negative signs of the coefficients of morning and night slots. During holidays, users prefer to receive parcels in the morning, but not in the evening. A possible explanation is that they may want to leave home after receiving the parcels or likely return home late in the holidays. As for occupation, self-employed people or people who have part-time or no jobs clearly do not opt for the night slot.

6.2.2. Error components

The standard deviations of the heteroskedastic error components are all significantly different from zero, except for the one for the normal delivery option. This suggests that there exists unobserved heterogeneity and significant substitution patterns associated with choices of delivery options, dates, and time slots. First, the difference in the scales of the error components of next-day, scheduled, and normal delivery indicates the heteroskedasticity among the option alternatives. In addition, the estimation results of the error scales of the day and slot alternatives support the cross-nested structure of Fig. 3, and alternatives within a common nest are correlated with each other.

6.3. Willingness-to-pay for delivery

Finally, we discuss the distributions of WTPs for delivery obtained from the estimated model. In the final model, all the distributional parameters of VODT, VOTS, and the sensitivity to the delivery fee were estimated with statistical significance. The polynomial approximation enabled estimating their distributions in a data-oriented manner, without imposing a parametric distributional form a priori. The estimation results of γ_{PI} and γ_{FR}, capturing the observed heterogeneity in the WTP distributions, were consistent with our expectations. The negative sign of the power γ_{PI} of the item price suggests that customers who order more expensive items are less sensitive to the delivery fee. The positive sign of the power γ_{FR} of the user's online shopping frequency indicates that the scale of VODT increases according to the frequency. Both results are reasonable and consistent with our hypothesis.

6.3.1. Value of delivery time savings

The distribution of VODT obtained from the estimation of MXL 4 is shown in Fig. 4, and its summary statistics are reported in Table 6.

Table 5
Estimation results.

	MNL		MXL 1		MXL 4	
	Estimate	Rob.std.err.	Estimate	Rob.std.err.	Estimate	Rob.std.err.
Delivery attributes						
Delivery fee (JPY)						
Fixed parameter	-0.009***	0.000	-0.018***	0.000		
Random parameter						
Lower bound (log)					-2.232***	0.078
Spread (log)					-5.036***	0.052
Observed heterogeneity						
Item price (JPY)	-0.106***	0.014	-0.089***	0.016	-0.069***	0.026
Value of delivery time savings (JPY/day)						
Fixed parameter	44.870***	1.724	26.465***	1.158		
Random parameter						
$\eta_{d,0}$					243.651***	0.992
$\eta_{d,1}$					-1,034.516***	1.175
$\eta_{d,2}$					1,061.363***	2.815
$\eta_{d,3}$					618.977***	2.384
$\eta_{d,4}$					-936.408***	2.250
Observed heterogeneity						
Online shopping frequency	0.104***	0.020	0.146***	0.033	0.160***	0.033
Value of time slot shortening (JPY/hr)						
Fixed parameter	9.412***	2.275	5.455***	2.088		
Random parameter						
$\eta_{h,0}$					31.248***	7.596
$\eta_{h,1}$					-120.229***	39.598
$\eta_{h,2}$					100.399**	42.696
Alternative specific constant						
Option choice (ref. = Normal delivery)						
Next-day delivery						
Baseline	-0.593***	0.067	-0.479***	0.102	-1.046***	0.142
Age ≤ 29 years old	0.479***	0.181	0.742**	0.297	1.254***	0.463
Age ≥ 70 years old	-0.373***	0.114	-0.569***	0.193	-0.611**	0.249
Male	0.177***	0.058	0.297***	0.099	0.175	0.141
Resident in Tokyo	-0.028	0.064	-0.096	0.109	-0.128	0.166
Free membership	0.415***	0.059	0.751***	0.102	0.767***	0.141
Telecommuter	0.102	0.104	0.168	0.176	0.402	0.275
Delivery box	0.157**	0.065	0.331***	0.112	0.436***	0.165
Scheduled delivery						
Baseline	-1.941***	0.120	-5.740***	0.302	-6.889***	0.571
Age ≤ 29 years old	0.238	0.210	0.206	0.406	0.833	0.567
Age ≥ 70 years old	-0.261**	0.129	-0.289	0.241	-0.932**	0.369
Male	-0.070	0.068	-0.332**	0.140	-0.562**	0.224
Resident in Tokyo	0.153**	0.071	0.259*	0.145	0.610**	0.277
Free membership	0.140**	0.067	0.018	0.137	-0.060	0.240
Telecommuter	-0.308**	0.123	-0.670***	0.237	-0.824*	0.494
Delivery box	-0.494***	0.078	-1.091***	0.160	-1.284***	0.285
Day choice (ref. = Weekday)						
Holiday						
Baseline	0.333***	0.092	3.063***	0.252	4.651***	0.450
Slot choice (ref. = Daytime slots)						
Morning slot						
Baseline	0.191	0.138	0.077	0.312	-0.159	0.511
Live with children	-0.110	0.097	-0.247	0.220	-0.216	0.384
Live with family	-0.219	0.166	-0.359	0.377	-0.747	1.097
Couple with double income	0.213*	0.116	0.463*	0.272	0.496	0.411
Live alone	0.348***	0.111	0.646**	0.265	0.602	0.530
Live in a detached house	-0.270***	0.078	-0.485***	0.176	-0.550	0.347
Holiday	0.411***	0.115	0.815***	0.176	0.723***	0.243
Public officer	-0.051	0.165	-0.009	0.384	0.244	0.694
Professional job	-0.053	0.212	-0.420	0.404	-0.526	0.954
Self employed	-0.144	0.167	-0.391	0.378	-0.053	0.578
Part-time job	-0.130	0.112	-0.426*	0.258	-0.347	0.358
No job	0.095	0.097	0.205	0.221	0.306	0.403
Night slot						
Baseline	0.520***	0.142	1.169***	0.264	1.765***	0.378
Live with children	0.034	0.113	0.186	0.208	0.246	0.271
Live with family	-0.731***	0.211	-1.127***	0.373	-1.162**	0.539
Couple with double income	0.172	0.134	0.528*	0.280	0.379	0.307
Live alone	-0.128	0.140	-0.252	0.255	-0.475	0.318
Live in a detached house	-0.168*	0.090	-0.289*	0.172	-0.434**	0.193
Holiday	0.028	0.113	-0.314**	0.157	-0.405**	0.203
Public officer	0.179	0.171	0.487	0.305	0.865*	0.511
Professional job	0.438**	0.201	0.857**	0.404	0.220	0.625
Self employed	-0.629***	0.239	-1.174***	0.452	-0.982**	0.488

(*continued on next page*)

Table 5 (*continued*)

	MNL		MXL 1		MXL 4	
	Estimate	Rob.std.err.	Estimate	Rob.std.err.	Estimate	Rob.std.err.
Part-time job	-0.257**	0.127	-0.619***	0.238	-0.768***	0.272
No job	-0.807***	0.127	-1.413***	0.263	-1.649***	0.316
Scales of error components						
Option choice						
Next-day delivery			2.235***	0.076	-2.082***	0.122
Scheduled delivery			-0.713***	0.197	2.108***	0.530
Normal delivery			0.055	0.226	-0.239*	0.143
Day choice						
Holiday			2.953***	0.144	3.588***	0.343
Weekday			3.652***	0.208	5.474***	0.414
Slot choice						
Morning slot			3.250***	0.161	-4.241***	0.218
Night slot			2.628***	0.170	-2.231***	0.328
Daytime slots			2.545***	0.195	2.400***	0.259

Table 6
Distribution characteristics of VODT.

	Mean	Std	Min	25%	50%	75%	Max	$P(w_d < 0)$
VODT (JPY/day)	44.459	43.308	-47.937	18.29	25.608	60.226	219.445	0.036

The VODT ranges from −47.937 to 219.445 JPY/day,[10] with 3.6% of users having negative values for delivery time savings. The median is 25.608 JPY/day, less than 10% of shipping fees that were used in the survey and are often adopted in Japanese delivery services.[11] This result implies that, in the delivery timing choice context, some customers do not necessarily need fast delivery, and rather prefer delivery as per their convenience, given their schedule constraints. Nevertheless, we note that some customers would highly value saving the delivery time, as the maximum VODT is 219.445 JPY/day. Therefore, there is a large variety of VODT in the population.

While VODT was analyzed in a single market in the literature, we can analyze the variation of VODT across different categories in a unified framework because our survey did not impose any restriction on the item category. Based on the estimation result, we performed an ex-post segmentation analysis of VODT,[12] and the results in Fig. 5(a) show its heterogeneity across different categories of ordered items. For software and computer-related devices, VODT indicates the highest values, followed by CD/DVDs, toys, and electric devices. A possible explanation is that these items are not readily available in nearby stores, but customers often want them urgently, thus requesting fast delivery. In fact, groceries have a lower VODT value than these items, because groceries for daily use can be obtained in nearby supermarkets. Moreover, customers ordering books and cosmetics have low VODT values, suggesting that such items are relatively inexpensive and are often not needed immediately. This is also seen in the result of Fig. 5(b), which shows that the more expensive the ordered item, the higher the VODT. Finally, Fig. 5(c) clearly shows that customers who frequently shop online have higher VODT values, that is, frequent customers need fast delivery more than low-frequent users.

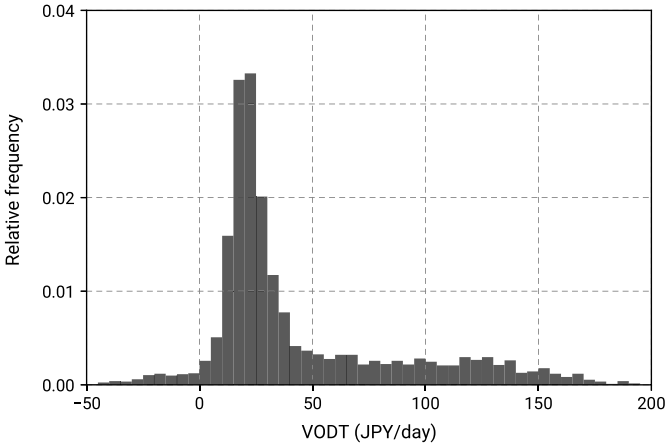


Fig. 4. Distribution of VODT among respondents.

6.3.2. *Value of time slot shortening*

The distribution of VOTS obtained from MXL 4 is shown in Fig. 6, and its statistics are reported in Table 7. The VOTS ranges from −3.827 to 27.086 JPY/hour, with 4.2% of the respondents having negative values. The median was 4.968 JPY/hour. As such, although a majority of customers would be willing to pay for the shortening of the delivery time slot size, the payment would be small. This means that customers do not highly value the reduction in time slot size in monetary terms.[13]

7. Concluding remarks

This study analyzed e-commerce customers' preferences for home delivery timing. We performed a stated choice survey, where customers were asked to indicate which option to choose among next, scheduled, and normal delivery, and if they chose scheduled delivery, they

[10] The average rate over 10 years of 2012–2021 is 106.1 JPY/$. Source: International Monetary Fund (IMF) Data https://data.imf.org/.

[11] For instance, as mentioned earlier, Amazon.co.jp set the shipping fee at 410–650 JPY for the goods sold by them.

[12] Because the estimation of an MXL model with a large sample and many parameters is very time-consuming, ex-post segmentation that only needs the estimation of a single global model is much more efficient than a priori segmentation (Hössinger et al., 2020). As such, ex-post segmentation allows the analyst to test various types of segmentation, and Hössinger et al. (2020) validated its usefulness and stability.

[13] Note that in our survey the time slot size ranged from two to four hours, and a two-hour slot might not have been sufficiently tight to increase the convenience of experienced e-commerce users.

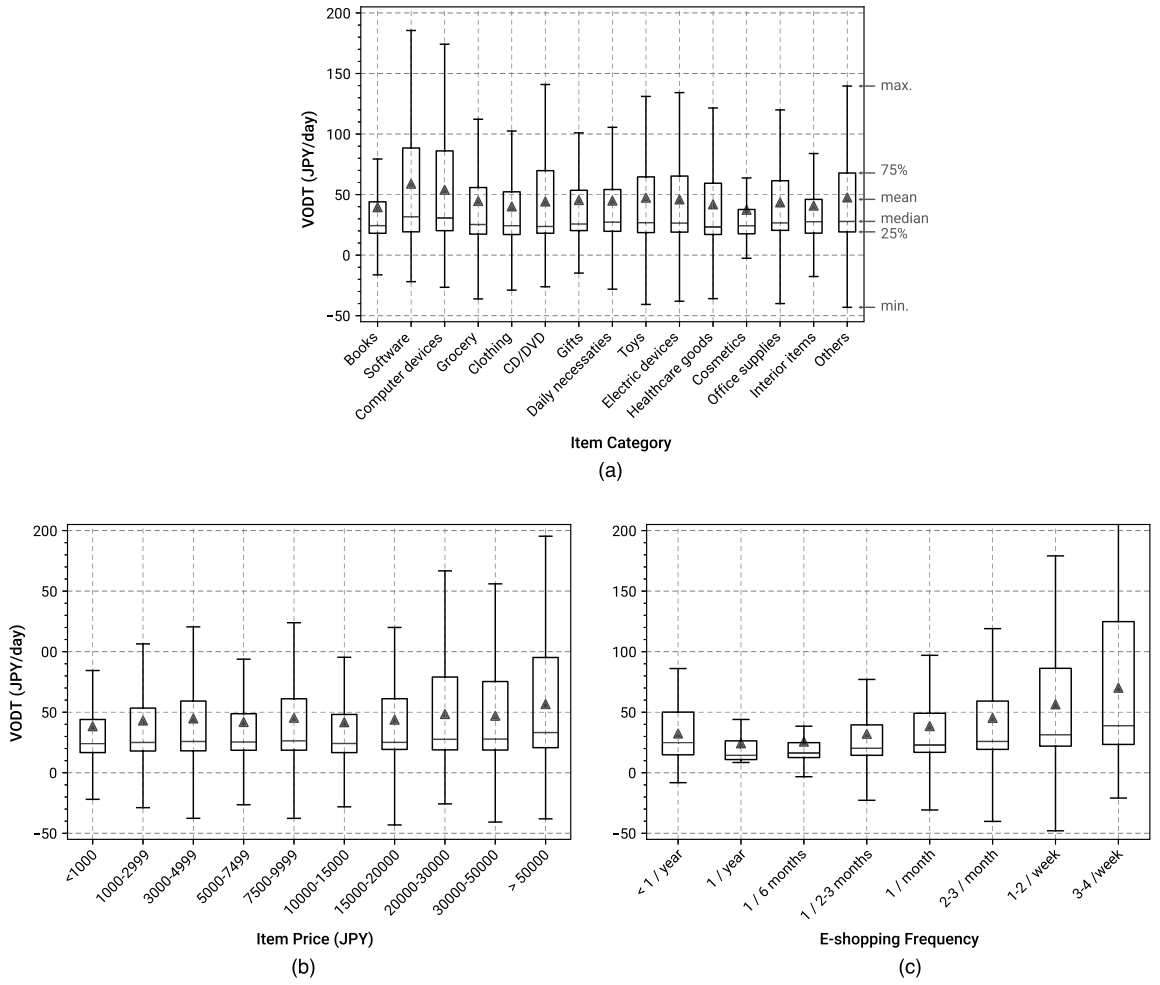


Fig. 5. VODT for different segments: focusing on (a) ordered item category, (b) ordered item price, and (c) e-shopping frequency of users.

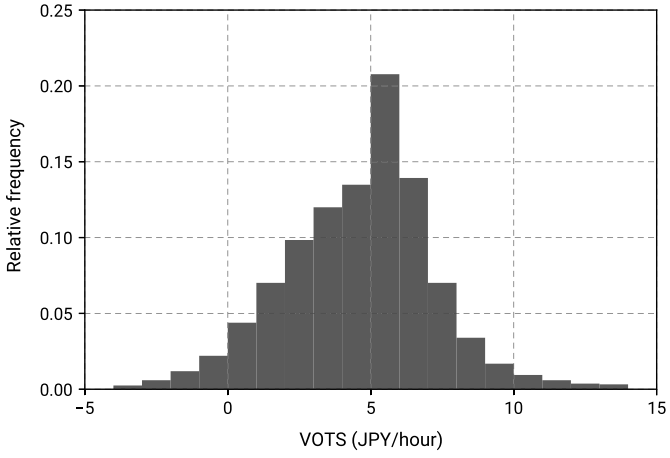


Fig. 6. Distribution of VOTS among respondents.

were also asked to select the delivery date and time slot. The stated choice data of 4,062 customers living in the three major metropolitan areas of Japan were analyzed by estimating a mixed logit model, capturing customers' heterogeneity and substitution patterns. We also applied a semi-nonparametric approach by Fosgerau and Mabit (2013) to flexibly estimate the distributions of WTP for delivery attribute changes.

7.1. Main results

The results of this study contribute to advancing the understanding of e-commerce customers' choice behavior of home delivery timing, which plays a key role in designing delivery demand management and retailing strategies (Neslin et al., 2006; Holguín-Veras et al., 2017). Specifically, the present results suggested that delivery service attributes including fee, time (speed), and time slot size are significant determinants of delivery timing choice. We found that elderly people do not tend to choose customer-oriented delivery options such as next-day and scheduled delivery, which complements the study of (Kvalsvik, 2022). The frequency of telecommuting and the presence of a delivery box also have a strong relationship with a low propensity to request scheduled delivery. However, customers who have a delivery box at home tend to choose next-day delivery, that is, although installing a delivery box reduces the demand for scheduled delivery and the risk of delivery failure, it may lead to an increased demand for fast delivery.

The analysis also revealed that the VODT is significantly heterogeneous among customers. Nevertheless, the median VODT was only 25.6 JPY per day, debunking the myth that everyone needs fast delivery and adding to the statement and results of Rai et al. (2019). In terms of heterogeneity, VODT had a high value for customers who frequently shop online and/or order expensive items, and varied according to the category of the item ordered. Additionally, the VOTS was found to be low and distributed with a median of 5.0 JPY per hour, implying that cus-

Table 7
Distribution characteristics of VOTS.

	Mean	Std	Min	25%	50%	75%	Max	$P(w_h < 0)$
VOTS (JPY/hour)	4.716	2.792	-3.827	2.979	4.968	6.216	27.086	0.042

tomers do not highly value the reduction in time slot size in monetary terms.

7.2. Managerial implications

Today many e-commerce marketplaces offer their customers fast delivery for free, which imposes a strict lead-time constraint on the delivery process. However, the large heterogeneity in VODT suggests a possibility for retailers to design delivery time and fees depending on customers or situations. The segmentation results of VODT particularly suggest the effectiveness of differentiated pricing by categories and prices of ordered items or e-shopping frequency. Moreover, the small median value of VODT implies that most customers are not willing to pay even 10% of the delivery fee to save the waiting time by one day. Therefore, an incentive (e.g., pricing or discounting) for customers to shift their choices to flexible delivery timing options can be small, which is good news for retailers and couriers.

Also, because the size of a time slot is a strict time-window constraint on the supply side for last-mile delivery route planning (e.g., Campbell and Savelsbergh, 2005; Agatz et al., 2011), the result on the estimates of VOTS may suggest that lengthening the size of delivery time slots would be a way to significantly improve routing efficiency without a serious reduction in user satisfaction.

Moreover, our WTP measures were calculated purely with respect to the delivery fee, not including item price (Gawor and Hoberg, 2019) or travel cost (Hsiao, 2009; Schmid and Axhausen, 2019; Meister et al., 2023). As such, the results of this study can be used for the level-of-service design for last-mile delivery, independent of the retailer/marketplace strategy.

7.3. Limitations and future work

It should be noted that the results of this study might be of limited value in demand forecasting in realistic settings because a stated choice experiment is likely subject to a hypothetical bias. The aggregation of time slot alternatives in the modeling process also might affect market share prediction.[14] This can be especially problematic when the time slot size is small and one aims to accurately forecast daytime demand. Nevertheless, considering the difficulty in observing the revealed preferences in a real delivery option choice situation, the results of this study potentially contribute to demand forecasting and the design of retailing and delivery demand management strategies, which we consider an important future work.

Since our model describes the integrated choices of option, date and time slot for delivery, it would be possible to analyze both day-to-day and within-day dynamics of delivery demand and their impact on routing efficiency, combined with a multi-period vehicle routing problem (Archetti et al., 2015). As such, future work includes a study of delivery attribute designs (e.g., pricing, lead-time and time-window designs) to achieve the operational efficiency of last-mile delivery while maintaining customer satisfaction.

While this study focused on the quantitative analysis of VODT and VOTS based on the RUM approach, a general marketing model of online consumer behavior (Laudon and Traver, 2023) suggests the importance of psychological factors. Current e-marketplaces often set customer-oriented delivery services (e.g., fast delivery) as default, and the nudge theory (Thaler and Sunstein, 2009) can be also effective in reasonably changing customer choice of delivery timing. The tracking of clickstream behavior (Laudon and Traver, 2023) and eye-gaze (Modi and Singh, 2023) can contribute to the further understanding of e-commerce consumer choice behavior. As such, future work can extend the framework beyond rationality.

In addition, as this study was carried out during the COVID-19 pandemic, under the declaration of a state of emergency, the data may not be typical of the life back to the normal. However, as telecommuting has penetrated the society,[15] the customers' behavior might not be very different between during and after the pandemic. Therefore, the continuation of the survey and the temporal comparison of customers' preferences for home delivery timing would be another important future work.

CRediT authorship contribution statement

Yuki Oyama: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Daisuke Fukuda:** Formal analysis, Methodology, Software, Writing – review & editing. **Naoto Imura:** Investigation, Project administration, Resources. **Katsuhiro Nishinari:** Investigation, Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Screen image of the stated choice survey

An example of the original version of a stated choice task presented to a respondent is shown in Fig. A.7, while Fig. 1 was its English translation.

Appendix B. VODT for different customer segments

Fig. B.8 shows the VODT estimates categorized by age, gender, and residence of the customers. While we did not observe a large difference with respect to age and residence, we found that the VODT of males is greater than that of females.

[14] Its effect on the VODT and VOTS estimates is limited because the aggregation does not change the attributes of the alternatives.

[15] Source: Mobile Society Research Institute "Telework implementation rate has remained flat over the past three years at about 15%: In Tokyo, about 30% of employees telework (translated by DeepL)" https://www.moba-ken.jp/project/lifestyle/20230925.html.

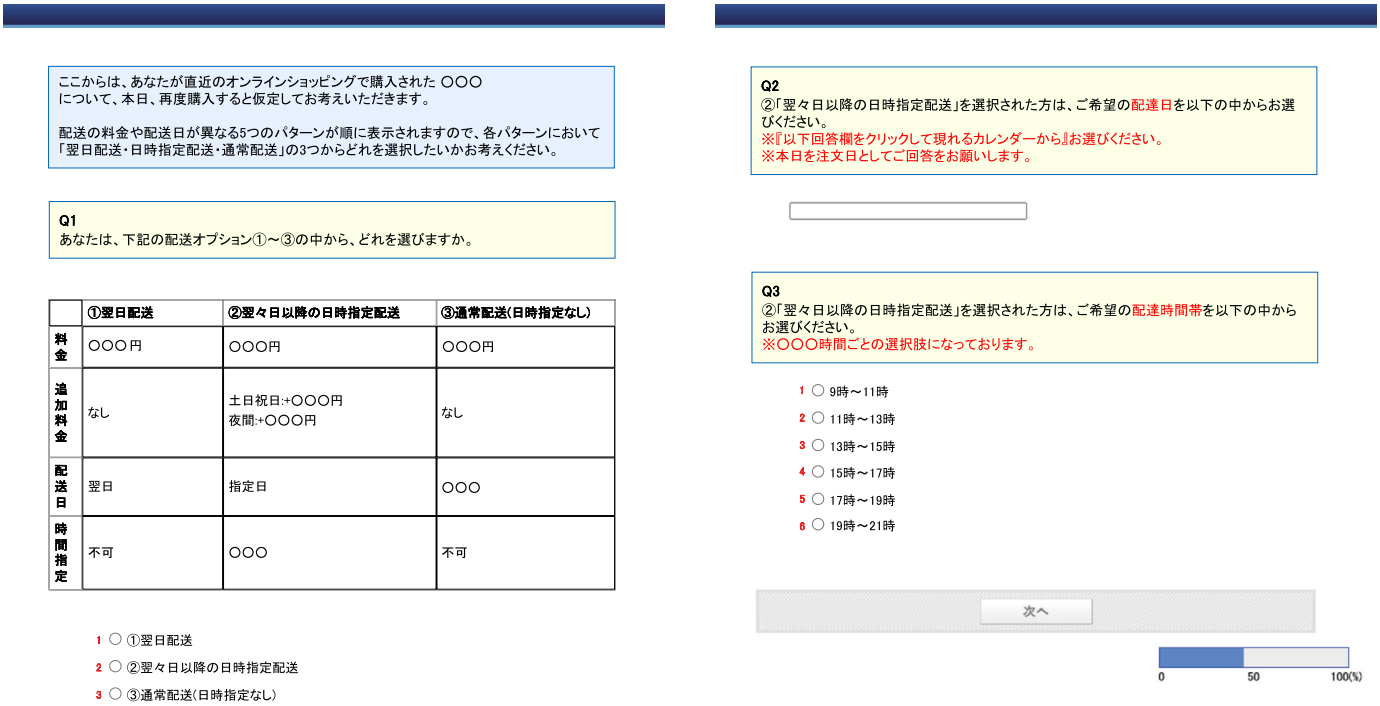


Fig. A.7. Screen image of a stated choice task (in Japanese). The three circles "○○○" in the blue introductory box indicate the item that the respondent ordered in the recent e-shopping transaction, and those in the table of Q1 are the attributes of the choice task.

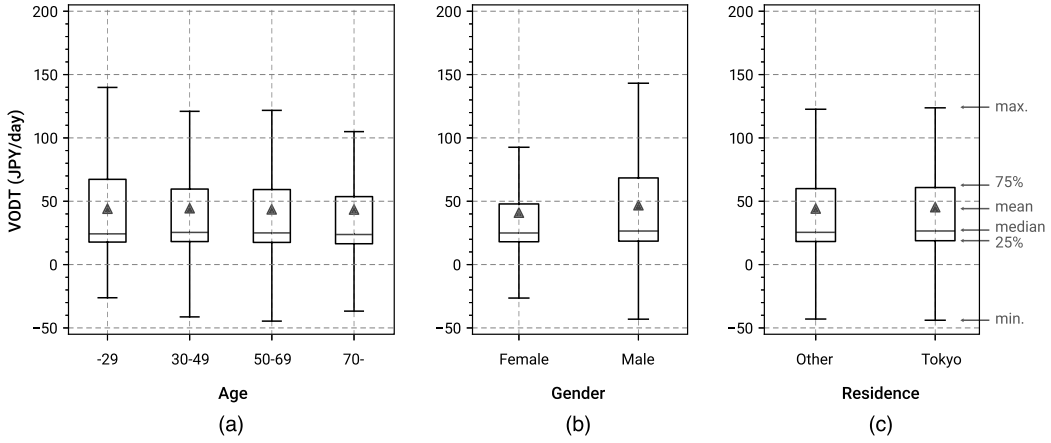


Fig. B.8. VODT for different customer segments by demographic characteristics: (a) age, (b) gender, and (c) residence (Tokyo or not).

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